(1) Your name and student ID

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(2) How to compile and execute your program and give an execution example.

先去 HW4/src,打 make

然後去 HW4/bin

./bin/hw4 <input.txt> <output.out>

Ex: ./bin/hw4 ../testcase/public1.txt ../output/public1.out

(3) The area and the runtime of each testcase, respectively. Paste the screenshot of the result of running the HW4_grading.sh.

```
checking item              | status
---------------------------|--------
correct tar.gz             | yes
correct file structure     | yes
have README                | yes
have Makefile              | yes
correct make clean         | yes
correct make               | yes

 testcase |          area |     runtime | status
----------|---------------|-------------|--------
  public1 |      48439944 |        3.85 | success
  public2 |        615072 |       50.95 | success
  public3 |        611861 |       86.77 | success
+-----------------------------------------------+
|                                               |
|   Successfully write grades to HW4_grade.csv  |
|                                               |
+-----------------------------------------------+
```

(4) The details of your implementation.

不同於講義上是利用 Hierarchical node 來處理，我是直接把一個 symmetry group(以下簡稱 group)包起來，用 B*tree 來進行優化。

主要分成兩階段，一開始只針對 group 裡面進行 simulated annealing，為了要保持對稱，方法便是只拿右半邊來做 sa，得到結果後再用鏡像的方式讓左右兩便會是對稱的情況，找到每個 group 最小的面積後，包起來且記住相對位置，再來進行所有 block 的 simulated annealing，找出所有 block 的最小面積。

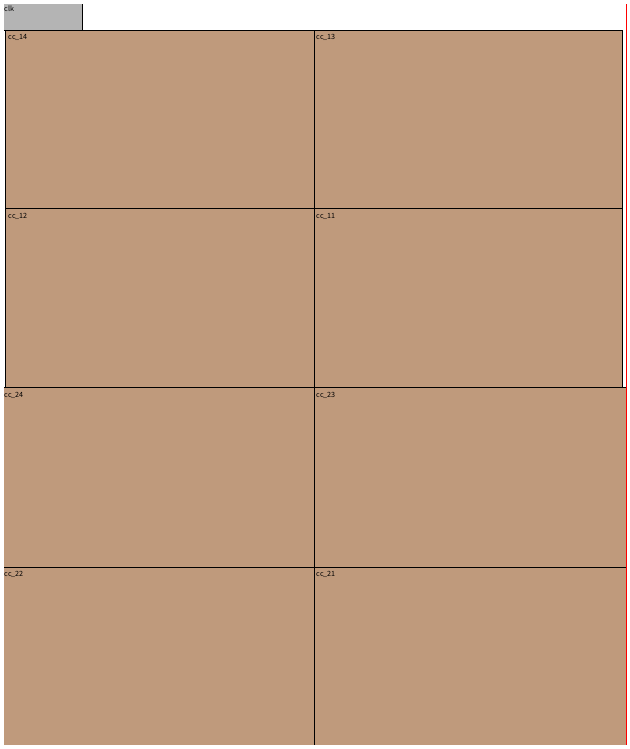
(5) Please describe your method of your initial placement.

對於每個 group 的 initial placement，我一律是採用全部疊起來的方法，因為我想說 node 數量偏少，所以不用特別排。

至於全部的 block 的 initial placement，我是採用跟之前一樣，先把 block 按照 height 進行排列，然後再一個一個排成一排，如果超過 outline 就往上排，這個 outline 也是跟前一個作業一樣設計，就是 floor(sqrt(AllCellArea))。

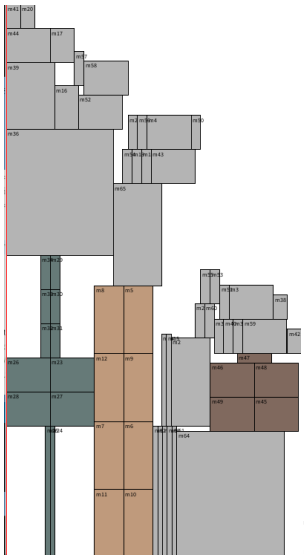
(6) What tricks did you do to speed up your program or to enhance your solution quality? Please use HW4_printer to generate the image to compare different stages (e.g., initial placement → trick 1 → trick 2 → final result) of your placement results.

Public1(幾乎沒有動)

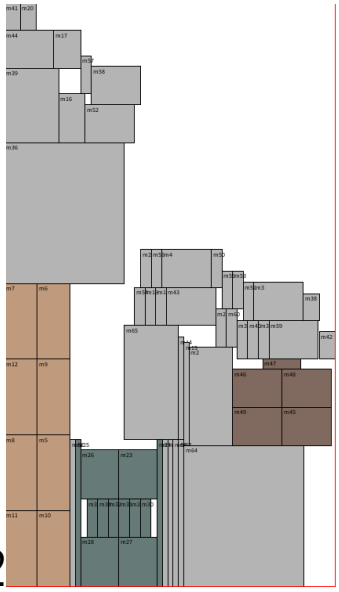


Public2

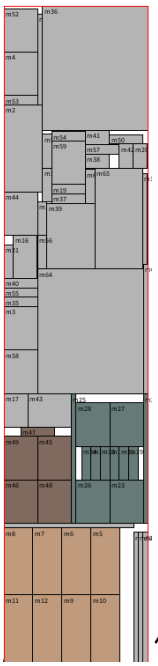
Initial 完，可以看到有很多空間被浪費。(1)



1->2

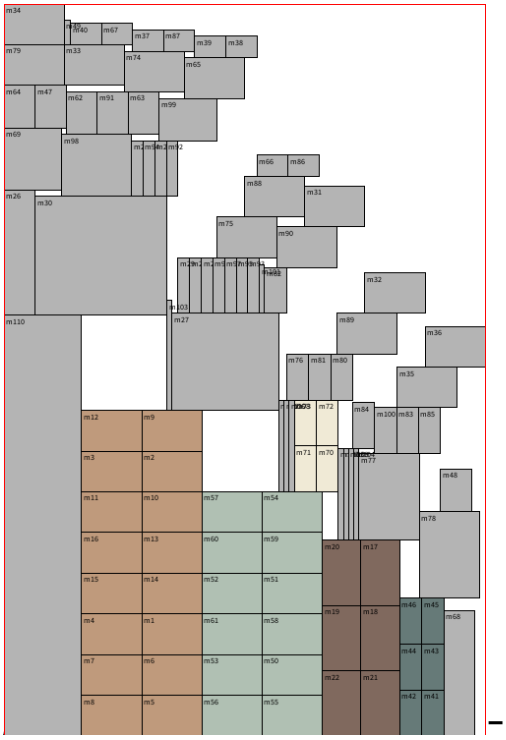


下一步可以看到 group 的面積有明顯變小。(2)

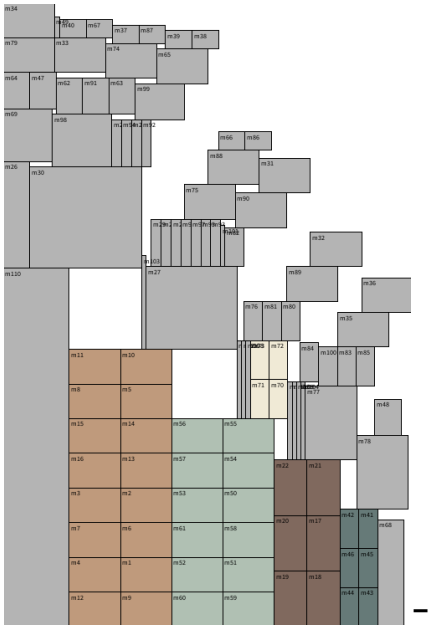


-> 做最後的 sa 得到結果。

Public3(流程與 public2 一樣)



->



->

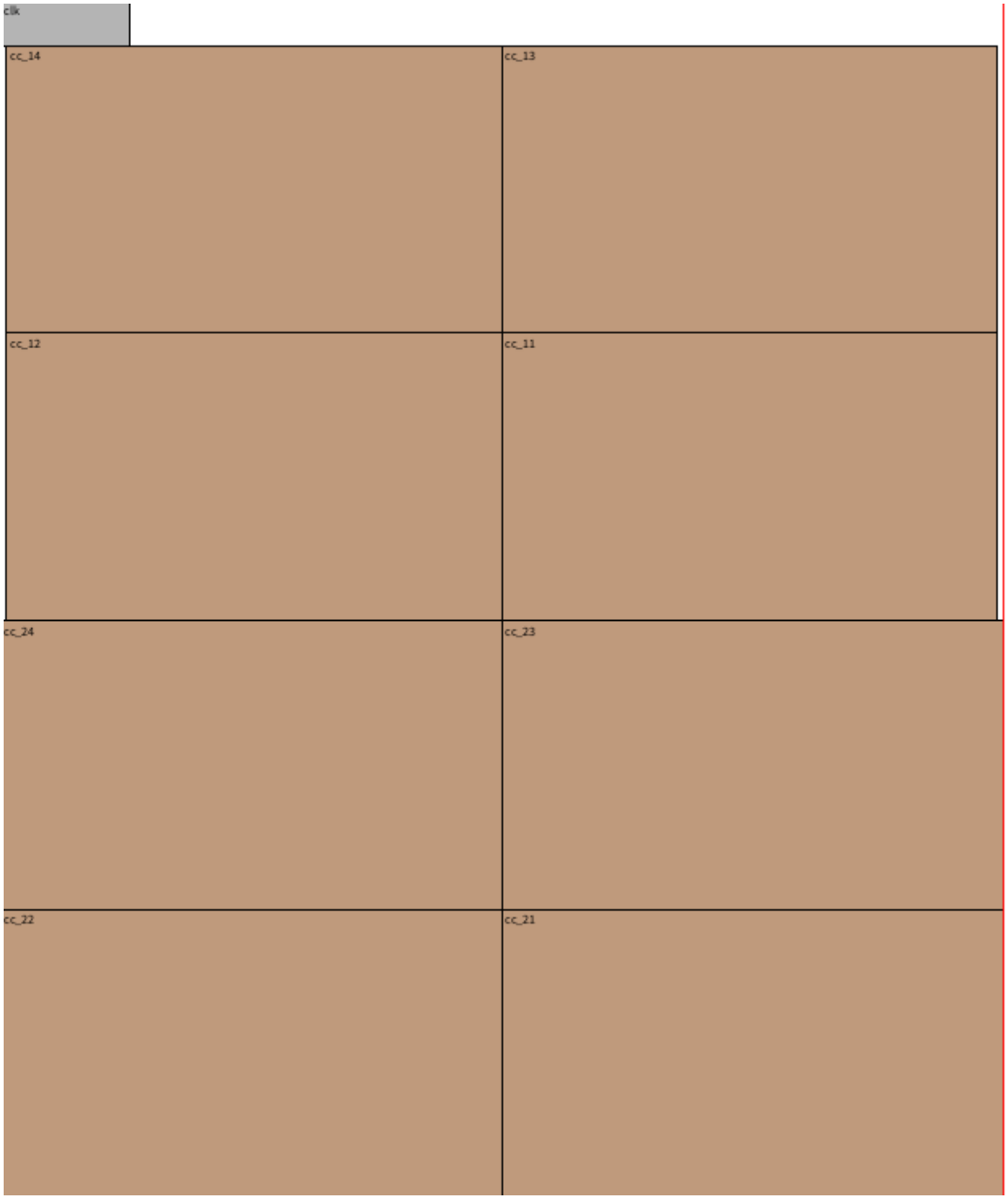
結果

(7) What have you learned from this homework? What problem(s) have you encountered in this homework?

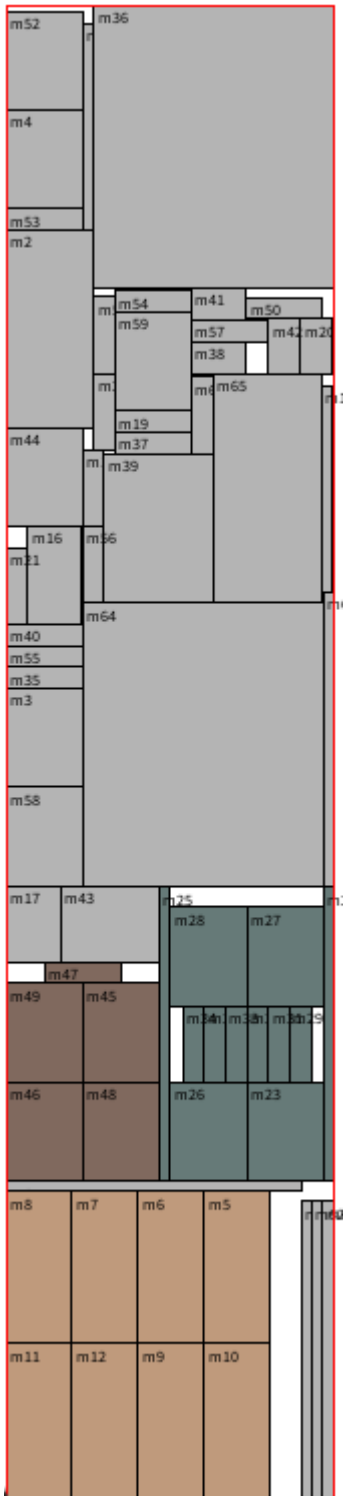
一開始我絞盡腦汁想要寫出講義上的方法，但對於原本的 B*tree 要擴充成特別的形式，像是 Hierarchical node 以及 contour node，但一直卡住，最後才退而求其次選這個方法，但這個方法就是不能避免會浪費一些空間，不過有完成作業就是值得慶幸的事....，希望有一天學會要怎麼寫講義上的方法。另外我有發現，因為我是做兩次 sa，第一次得到面積最小的結果，不一定會在第二次 sa 得到最優的結果，所以要多測試一些不同情況才可以得到最優解。

Result

Public1.txt



Public2.txt


m52
m36
m4
m53
m2
m54
m41
m50
m59
m57
m42
m38
m65
m19
m37
m44
m39
m16
m21
m64
m40
m55
m35
m3
m58
m17
m43
m28
m27
m47
m49
m45
m46
m48
m26
m23
m8
m7
m6
m5
m11
m12
m9
m10

Public3.txt

m27
m106
m88
m79
m109
m77
m107
m78
m39
m104
m38
m66
m83
m37
m96
m30
m69
m95
m90
m26
m32
m31
m71
m102
m70
m48
m65
m76
m73
m72
m89
m87
m110
m62
m34
m108
m94
m84
m74
m103
m25
m22
m47
m98
m64
m82
m63
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m80
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m13
m75
m12
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m86
m68
m81
m91
m97
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m6
m53
m50
m100
m67
m92
m4
m1
m60
m59
m85
m20
m17
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m45
m19
m18
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m52
m51
m44
m43
m11
m10
m61
m58
m22
m21
m42
m41
m3
m2
m56
m55